

A Reader's Map to the Event-Density Gravity Line

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Seven documents (the first five openly archived on Zenodo — each link opens that record; GR-III and GR-IV are in the repository, archive pending):

Shorthand	Title	Archived
GR-I (<i>General Relativity I</i>)	The Bandwidth Lapse and the Factor of Two	zenodo.org/records/20665981
GR-II (<i>General Relativity II</i>)	The Arrow's Fingerprint	zenodo.org/records/20666023
GR-III (<i>General Relativity III</i>)	The Arrow's Engine — the dynamical rule built	in the repository (archive pending)
GR-IV (<i>General Relativity IV</i>)	The Arrow's Alibi — the preferred-frame front, closed	in the repository (archive pending)
KM-I (<i>Khronon-MOND I</i>)	The Arrow's Deep Field	zenodo.org/records/20672818
KM-II (<i>Khronon-MOND II</i>)	The Arrow's Horizon	zenodo.org/records/20672870
One Field (<i>the capstone letter; the others summarized</i>)	Gravity, Dark Matter, Dark Energy, and the Arrow of Time in Event Density	zenodo.org/records/20678990

Together they are the mechanism by which Event Density (ED) — a discrete, relational substrate with no fundamental spacetime — climbs to a theory that looks and behaves like General Relativity.

But the real story is richer than “an emergent-gravity model.” ED does not approximate GR; it lands exactly on the Einstein weak field, then shows precisely where and why it deviates — and the deviation is the arrow of time's fingerprint.

The gravitational class it lands in is called khronometric gravity (the infrared limit of Hořava gravity) — a theory physicists already study.

What ED adds is not standard: the khronon is not postulated but counted, it is precisely the arrow of time made dynamical, and the same single field goes on to carry the dark-matter phenomenology and the dark-energy scale. **One field, one scale, five roles.**

Below is the structural map of how the seven documents assemble the bridge:

ED → **Einstein weak field** → **khronometric gravity** → **MOND** → **cosmology**
→ Λ .

1. GR-I: The Metric Emerges (and it's Einstein's)

Takeaway: ED's bandwidth field b is the spatial metric, and the commitment rate fixes the lapse.

Result:

- $g_{ij} \sim b^{-1}$
- $N^2 \sim b$
- $\Rightarrow g_{00}g_{rr} = -1$ (the Schwarzschild relation)
- $\Rightarrow$ Einstein's weak-field metric, including the factor-of-two light bending.

This is the reason ED can become a metric theory at all. The arrow is already present here — the lapse is fixed by the commitment rate (primitive P11, irreversibility) — but it does not yet show up as a deviation from Einstein, because khronometric gravity coincides with GR in the static weak field. The fingerprint surfaces in GR-II.

2. GR-II: The Arrow Forces the Gravitational Class

Takeaway: ED is not pure GR. The arrow lives in the law, so the gauge group is reduced.

Result:

- Gauge: foliation-preserving diffeomorphisms
- Modes: 2 tensor + 1 scalar
- Class: khronometric gravity (Hořava IR limit)

Structural passes:

- $c_t = c$ (GW170817 satisfied as an identity)
- universal Lorentz violation (safe from 10^{-20} bounds)

This is where ED becomes Einstein + one scalar — the scalar is the arrow made dynamical. GR-II names the preferred-frame parameters α_1, α_2 as the falsification target and defers them; GR-III/GR-IV resolve them.

3. GR-III: The Engine — the Dynamical Rule Built

Takeaway: The rule the field equation reduces to is built and runs; the weak-field assumptions GR-I/II carried are closed.

Result:

- Dynamical rule $\dot{b} = D\nabla^2 b - \kappa\rho$; steady state = the Newtonian field equation (corr. 0.999); $r_s \propto M$.
- A spontaneously forming $b \rightarrow 0$ horizon (a frozen decoupling cut) carrying area-law entropy $S \propto A$ and Hawking scaling $\kappa \propto 1/r_h$.
- $\alpha = 1$ forced (Einstein, not Nordström — the reserve cannot replenish, so the arrow selects the branch).
- The khronon speed pinned: $c_s = c$ (the reserve sector is dissipative, not kinetic).

This is the engine: one irreversibility (the arrow, P11) selects the Einstein branch, makes the khronon physical, and keeps it on the light cone. It builds the $c_s = c$ that GR-IV's preferred-frame reduction consumes.

4. GR-IV: The Alibi — the Preferred-Frame Front, Closed

Takeaway: The preferred-frame parameters that generically *kill* khronometric theories are, for ED, suppressed — by the same sparseness of becoming that gives ED quantum mechanics.

Result:

- $\alpha_2 = 0$ **exactly** — a structural consequence of both gravitational cones being luminal ($c_T = c$, $c_s = c$); the published khronometric formula vanishes identically on that surface (literature-verified). The *tighter* preferred-frame bound is met for free.
- $\alpha_1 = -4\lambda_{\text{local}}$, with $\lambda_{\text{local}} \sim \rho_{\text{event}}/\rho_{\text{Planck}}$ — the commitment sparsity, because the metric stiffness is the always-on sharing while the khronon stiffness is the sparse act of commitment.
- Commitment is *forced* sparse: dense commitment is the quantum Zeno limit (continuous measurement freezes all evolution $\rightarrow$ no quantum mechanics), and ED has a quantum sector. So $\alpha_1 \lesssim 4 \times 10^{-93}$ — safe below the bound by ≥ 70 orders.
- MOND is untouched: $a_0 = cH_0/2\pi$ is the khronon's *cosmological* face, not the local stiffness.

So the front is closed *favorably* — and stays a live test: a measured $\alpha_1 \gtrsim 10^{-5}$, or *any* nonzero α_2 , still kills the theory. **One structure, three faces:** sparse becoming gives ED quantum coherence, gravitational time dilation, and preferred-frame safety.

5. KM-I: The Deep Field = MOND

Takeaway: The khronon's deep-infrared regime is the MOND sector.

Result:

- $a_0 = cH_0/2\pi$ inherited from ED cosmology
- Deep-field law: $a = \sqrt{(a_N a_0)}$ (matched to the already-derived Combination Rule — forced-given-Paper_030, an embedding, not a derivation from primitives)
- Lensing passes with no vector field
- Solar system screened
- No ghosts (W -sector adds no time derivatives)

This is how ED unifies GR's weak field with MOND's deep field using one scalar field.

6. KM-II: The Cosmological Sector and the Dark Fluid

Takeaway: The same scalar has a cosmological face — a dark fluid whose sound speed is the free dial.

Result:

- Regulator family $\mathcal{W}(A^2, \Theta)$
- Orthogonality theorem: cosmology is sequestered from statics
- Dust-like clustering requires $c_s \approx 0$
- The near-vacuum soft spot = the CMB/cluster dial
- Late-time constant automatically $\sim H_0^2 c^2 / G$

This is the moment ED's scalar is positioned to serve as a dark-matter analogue on cosmological scales — addressable, not discharged: no dial member is chosen and no CMB spectrum is computed.

7. One Field, One Scale, One Λ

(the capstone letter; the Λ result itself lives in KM-II §7.1)

Takeaway: The khronon's vacuum term and the substrate V1 integral are the same Λ at two levels.

Result:

- Both faces scale as $H_0^2 c^2 / G$
- Sign check passes
- EFT constant pinned: $\mathcal{W}_0 = -24\pi^2 \Omega_\Lambda \approx -162$
- Substrate integral must hit that number $\rightarrow$ falsifiable

This is when ED's cosmology closes: Λ is not two things; it is one boundary seen two ways.

(The $\mathcal{W}_0$ derivation is carried in KM-II, and the letter restates it.)

8. The Big Picture: How ED Reaches GR

ED starts with a discrete substrate whose only special structure is an irreversible arrow of time.

That arrow forces a preferred foliation, which forces a khronon, which forces khronometric gravity, which coincides with Einstein in the weak field, and whose deep field and cosmological field reproduce MOND and Λ .

Or even shorter:

The arrow $\rightarrow$ the khronon $\rightarrow$ Einstein + MOND + Λ .

This is why the capstone letter says:

One field. One scale. Five roles. Three ways to kill it. One number to miss.

The One-Field paper is the natural place to start in reviewing the ED-GR connection. The whole claim is there in five pages.